

Thermo-elastic analysis of functionally graded graphene nanoplatelets (GPLs) reinforced closed cylindrical shells

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ABSTRACT

This paper investigates, for the first time, the thermo-elastic responses of a functionally graded (FG) graphene nanoplatelets (GPLs) reinforced composite (FG-GPLRC) closed cylindrical shell under a temperature field due to steady-state thermal conduction along thickness direction. A comprehensive thermo-elastic model covering various thermal boundary conditions (TBCs) is established. Analytical solutions for both radial stress and hoop stress of the shell are derived based on the plane hypothesis and displacement continuity conditions between the cylindrical shell and hemispherical ends. A parametric study is conducted to investigate the effects of GPL distribution pattern, weight fraction and geometry as well as TBCs on thermal stresses and deformation of the shell. It is found that GPL distributions V and O are preferred patterns for both FG-GPLRC cylindrical shell and hemispherical ends, respectively. The closed FG-X-GPLRC cylindrical shell with higher GPL concentration has higher thermal conductivity consequently higher thermal stresses. The research findings are of practical importance for the applications and design of closed FG-GPLRC shells in a variety of engineering sectors.

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1. Introduction

Hollow cylindrical shells have been widely used as critical structural components in aerospace, nuclear reactors, chemical plants, oil-gas storage facilities and transportation tools [1,2]. Owing to its low centrifugal forces, high strength and capacity, cylindrical shells are also applicable in the aircraft propulsion systems and rotary dryers [3–5]. These components are often exposed to high temperature, high fluid pressure, high rotating speed and electromagnetic environments [6–9]. Functionally graded materials (FGMs) which are microscopically inhomogeneous materials with physical properties varying smoothly and continuously from one surface to the other are introduced to improve their thermal and mechanical performance [10–13]. Chen and Lin [14] found that the property of FGMs had a significant influence to the stress distribution of thick cylinders and spherical pressure vessels along the radial direction. Wang [15] proposed an effective approach to analyze the effect of the inhomogeneity on the transient thermal responses of a functionally graded hollow cylinder. Burlayenko et al. [16] employed graded finite element method to study the heat transfer and thermal stress of FGMs. Dong et al. [17] presented an analytical study to predict the low-velocity impact response of a spinning functionally graded (FG) graphene reinforced cylindrical

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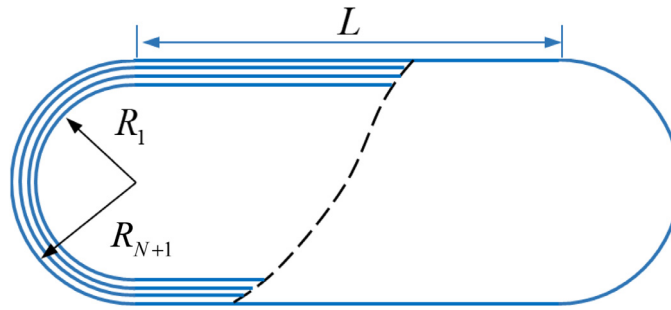


Fig. 1. A closed FG-GPLRC cylindrical shell.

shell subjected to impact, external axial and thermal loads. However, the above studies are limited to thermally induced responses of open functionally graded hollow cylinders only. In many cases, FGMs cylinders need to be sealed at both ends to sustain high pressure and/or high temperature but research on this type of shell structures is quite limited.

Graphene and its derivatives have become one of the most promising reinforcing nanofillers in composite structures due to their excellent mechanical, and thermal characteristics [18,19]. Previous studies have demonstrated both experimentally and numerically that GPLs considerably outperform other commonly used nanofillers such as carbon fibers (CFs) and carbon nanotubes (CNTs). Rafiee et al. [20] found that the addition of a very low content of graphene could significantly improve the tensile strength and stiffness of polymer composites. Liang et al. [21] observed a significant improvement in mechanical properties of graphene based nanocomposites when 0.7 wt.% of graphene oxide was added into the polymer matrix. Wan et al. [22] studied the fatigue mechanisms of thermosetting grapheme-based polymer nanocomposites (TGPNS) and found that the TGPNS effectively restricted the crack growth when graphene was used as reinforcing nanofillers. In order to make the best use of the significant reinforcing effects of GPLs on the mechanical properties of nanocomposites, a functionally graded GPLs-reinforced composite (FG-GPLRC) structure was introduced in which GPLs were non-uniformly and randomly dispersed in the polymer matrix [23–28]. Javani et al. and Kiani [29–31] investigated the nonlinear free vibration and thermal buckling of FG graphene platelet reinforced composite annular and laminated skew plates. Dong et al. [32,33] analyzed the nonlinear free vibration and buckling behavior of FG-GPLRC cylindrical shell and discussed the effect of spinning motion and axial load on the structural behaviors of the shell. Raheb et al. [34] examined the nonlinear stability and free vibration features of multilayer functionally graded graphene platelet-reinforced polymer composite (FG-GPLRPC) rectangular plates under compressive in-plane mechanical loads. Yang et al. [35–38] studied the nonlinear buckling, free and forced vibration characteristics of FG-GPLRC arches/beam under thermal, electrical and mechanical loadings. Their results showed that the structural behavior that was sensitive to both temperature and applied voltage could be remarkably improved by adding a low content of GPLs. Liu et al. [39] examined the three-dimensional buckling and free vibration characteristics of initially stressed FG-GPLRC cylindrical shells. By employing Lord-Shulman theory, Heydarpour et al. [40] analyzed the transient thermoelastic behavior of FG-GPLRC spherical shells under a thermo-mechanical loading and concluded that the addition of a very small amount of GPLs into polymer matrix not only significantly increased the shell stiffness and heat wave speed but also considerably changed the temperature and displacement distributions of the shell.

Thermal induced stresses are of vital importance for the safety and integrity of FG-GPLRC cylindrical shells under high temperature. However, no previous work has been done on the thermo-elastic behavior of closed FG-GPLRC cylindrical shell in thermal environment. To fill this research gap and address this important issue, this paper investigates the thermo-elastic responses of closed FG-GPLRC cylindrical shell based on a comprehensive thermo-elastic model capable of dealing with more thermal boundary conditions (TBCs) than the existing ones. The effective Young's modulus of each GPLRC layer are calculated by using modified Halpin-Tsai model while the other effective material properties are determined by the rule of mixture. The plane hypothesis and displacement continuity conditions between the cylindrical shell and two hemispherical ends are employed to derive the thermal stress and displacement solutions of the shell. The effects of GPL distribution pattern, weight fraction, geometry and dimensions as well as TBCs on the thermal stress and deformation of the shell are studied comprehensively.

2. Effective material properties

A closed FG-GPLRC cylindrical shell with length L , inner radius R_1 , outer radius R_{N+1} and with two hemispherical ends is shown in Fig. 1. It is assumed that the shell consists of multiple perfectly bonded isotropic homogeneous GPLRC layers in which each GPLRC layer is made from a mixture of polymer matrix and uniformly dispersed and randomly oriented GPLs. In addition, GPL concentration varies layer-by-layer along the thickness direction of the shell according to five different patterns (i.e., patterns U, X, O, V and A) as shown in Fig. 2 where a darker color indicates more GPLs are used. As a special case, GPL concentration is the same across all layers in Pattern U whereas Pattern O (Pattern X) represents a symmetric distribution with GPL content increasing (decreasing) from the inner and outer layers to the middle of the thickness. The unsymmetric

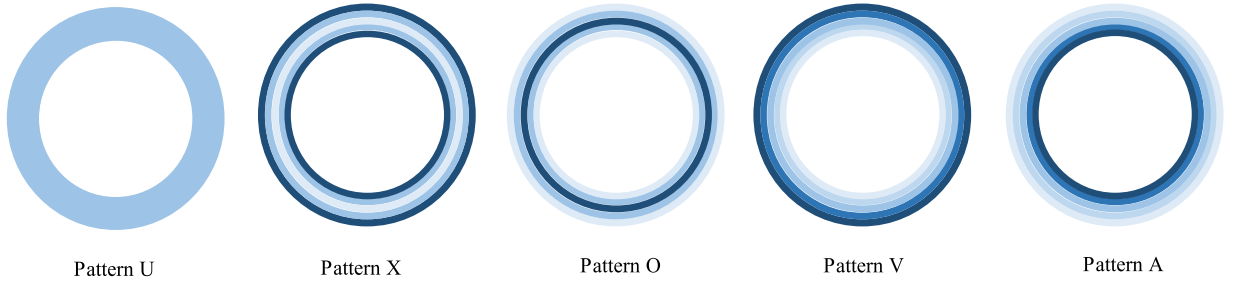


Fig. 2. Five GPLs distribution patterns.

Pattern A (Pattern V) exhibits a gradual increase (decrease) in GPL concentration from the outer layer to the inner layer of the shell.

For these GPL distribution patterns, the weight fraction of GPLs in the k -th layer $f_G^{(k)}$ can be expressed as

$$\begin{aligned}
 \text{Pattern U : } f_G^{(k)} &= f_G \\
 \text{Pattern X : } \begin{cases} f_G^{(k)} = 2f_G - \frac{4f_G}{N-2}(k-1)k \leq \frac{N}{2} \\ f_G^{(k)} = \left(k - \frac{N}{2} - 1\right) \frac{4f_G}{N-2} k > \frac{N}{2} \end{cases} \\
 \text{Pattern O : } \begin{cases} f_G^{(k)} = \frac{4f_G}{N-2}(k-1)k \leq \frac{N}{2} \\ f_G^{(k)} = 2f_G - \left(k - \frac{N}{2} - 1\right) \frac{4f_G}{N-2} k > \frac{N}{2} \end{cases} \\
 \text{Pattern V : } f_G^{(k)} &= 2f_G(k-1)/(N-1) \\
 \text{Pattern A : } f_G^{(k)} &= 2f_G(N-k)/(N-1)
 \end{aligned} \tag{1}$$

where N = the total number of layers, f_G = the total GPL weight fraction.

According to modified Halpin-Tsai model [41], the effective Young's modulus of each GPLRC layer is determined as

$$E_k = \frac{3}{8} \frac{1 + \xi_L \eta_L V_{Gk}}{1 - \eta_L V_{Gk}} \times E_M + \frac{5}{8} \frac{1 + \xi_W \eta_W V_{Gk}}{1 - \eta_W V_{Gk}} \times E_M \tag{2}$$

with

$$\eta_L = \frac{(E_G/E_M) - 1}{(E_G/E_M) + \xi_L}, \quad \eta_W = \frac{(E_G/E_M) - 1}{(E_G/E_M) + \xi_W} \tag{3}$$

$$\xi_L = 2l_G/h_G, \quad \xi_W = 2w_G/h_G. \tag{4}$$

in which E_G and E_M are the Young's moduli of GPLs and polymer matrix, respectively, V_{Gk} is the volume fraction of GPLs in the k th layer, l_G , h_G and w_G are the length, thickness and width of GPLs, respectively. The volume fraction V_{Gk} in the k th layer is

$$V_{Gk} = \frac{f_G^{(k)}}{f_G^{(k)} + (\rho_G/\rho_M)(1 - f_G^{(k)})} \tag{5}$$

where ρ_G and ρ_M are the mass densities of the GPL and polymer matrix, respectively.

The Poisson's ratio ν , thermal expansion coefficient α , thermal conductivity coefficient λ , and specific thermal capacity c of GPLRC can be determined by the rule of mixture as [42,43]

$$\begin{aligned}
 \nu_k &= \nu_G V_{Gk} + \nu_M (1 - V_{Gk}) \\
 \alpha_k &= \alpha_G V_{Gk} + \alpha_M (1 - V_{Gk}) \\
 \lambda_k &= \lambda_G V_{Gk} + \lambda_M (1 - V_{Gk}) \\
 c_k &= c_G V_{Gk} + c_M (1 - V_{Gk})
 \end{aligned} \tag{6}$$

with subscripts "G" and "M" refer to GPLs and polymer matrix, respectively. Moreover, the volume fractions V_{Mk} and V_{Gk} are related by $V_{Mk} = 1 - V_{Gk}$.

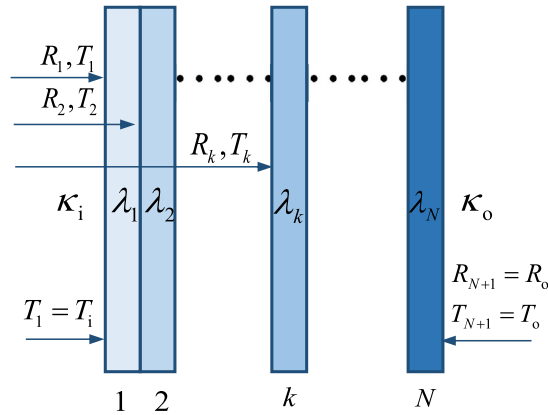


Fig. 3. Thermal transfer model of the FG-GPLRC cylindrical shell.

3. Governing equations

3.1. Thermo-elastic analysis for cylindrical shell

This section presents the thermo-elastic analysis of the cylindrical section in the closed FG-GPLRC cylindrical shell based on the thermal transfer model described in Fig. 3, in which T_k and λ_k ($k = 1, 2, 3, \dots, N$) are the temperature at each interface and thermal conductivity of each GPLRC layer, respectively. T_i , T_o , and κ_i , κ_o are the environmental temperature and coefficients of thermal convection at the inner and outer surface, respectively. The following four thermal boundary conditions (TBCs) are considered

- (1) TBC 1: The temperature is fixed at both inner and outer surfaces of the shell.
- (2) TBC 2: The temperature is fixed at the outer surface and the inner surface is in thermal convection.
- (3) TBC 3: The temperature is fixed at the inner surface is fixed and outer surface is in thermal convection.
- (4) TBC 4: Both the inner and outer surfaces are in thermal convection.

In the absence of heat generation, the steady-state thermal conductivity equation of the FG-GPLRC cylindrical shell is given as

$$\left(\frac{d^2}{dr^2} + \frac{1}{r} \frac{d}{dr} \right) T(r) = 0 \quad (7)$$

and its general solution is

$$T(r) = A_k \ln r + B_k (R_k \leq r \leq R_{k+1}, k = 1, 2, \dots, N) \quad (8)$$

where A_k and B_k are the constants determined from thermal boundary conditions. The FG-GPLRC cylindrical shell also satisfies the continuous thermal flux condition along the radial direction as

$$\lambda_k \frac{dT_k(r)}{dr} \bigg|_{r=R_{k+1}} = \lambda_{k+1} \frac{dT_{k+1}(r)}{dr} \bigg|_{r=R_{k+1}} \quad (9)$$

Combining Eqs. (8) and (9) gives

$$A_k \lambda_k = A_{k+1} \lambda_{k+1}, \quad \frac{T_k - T_{k+1}}{T_{k+1} - T_{k+2}} = \beta_k (k = 1, 2, \dots, N-1) \quad (10)$$

with

$$\beta_k = \frac{\lambda_{k+1} (\ln \frac{R_{k+1}}{R_k} + \omega_1 \frac{\lambda_k}{R_k \kappa_i})}{\lambda_k (\ln \frac{R_{k+2}}{R_{k+1}} + \omega_2 \frac{\lambda_{k+1}}{R_{k+2} \kappa_o})} \quad (11)$$

Hence, the temperature at each interface can be obtained as

$$\begin{bmatrix} 1 + \beta_1 & -\beta_1 & 0 & \cdots & 0 \\ -1 & 1 + \beta_2 & -\beta_2 & \ddots & 0 \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\beta_{N-2} \\ 0 & 0 & 0 & -1 & 1 + \beta_{N-1} \end{bmatrix} \begin{Bmatrix} T_2 \\ T_3 \\ \vdots \\ T_N \end{Bmatrix} = \begin{Bmatrix} T_1 \\ 0 \\ \vdots \\ 0 \\ \beta_{N-1} T_{N+1} \end{Bmatrix} \quad (12)$$

and A_k and B_k are determined as

$$\begin{cases} A_k = \frac{T_{k+1} - T_k}{\omega_1 \lambda_k / (R_k \kappa_i) + \ln(R_{k+1}/R_k) + \omega_2 \lambda_k / (R_{k+1} \kappa_o)} \\ B_k = \frac{\omega_1 \frac{\lambda_k}{R_k \kappa_i} T_{k+1} + \omega_2 \frac{\lambda_k}{R_{k+1} \kappa_o} T_k - T_{k+1} \ln R_k + T_k \ln R_{k+1}}{\omega_1 \lambda_k / (R_k \kappa_i) + \ln(R_{k+1}/R_k) + \omega_2 \lambda_k / (R_{k+1} \kappa_o)} \end{cases} \quad (k = 1, 2, \dots, N) \quad (13)$$

where ω_1 and ω_2 are constants distinguishing the given thermal boundary conditions. $\omega_1 = 0$ (or $\omega_2 = 0$) corresponds to the temperature at the inner (outer) surface being fixed. $\omega_1 = 1$ (or $\omega_2 = 1$) stands for thermal convection at the inner (outer) surface and in this case T_1 (or T_{N+1}) is to be replaced by T_i (or T_o).

According to thermo-elasticity theory, the thermal induced stress of cylindrical shell in plane strain state can be expressed as

$$\begin{cases} \sigma_r = \frac{E(1-\mu)}{(1+\mu)(1-2\mu)} \left(\varepsilon_r + \frac{\mu}{1-\mu} \varepsilon_\theta \right) - \frac{1+\mu}{1-\mu} \alpha T \\ \sigma_\theta = \frac{E(1-\mu)}{(1+\mu)(1-2\mu)} \left(\varepsilon_\theta + \frac{\mu}{1-\mu} \varepsilon_r \right) - \frac{1+\mu}{1-\mu} \alpha T \\ \tau_{r\theta} = 0 \end{cases} \quad (14)$$

Introducing the geometrical equation,

$$\varepsilon_r = \frac{du_r}{dr}, \varepsilon_\theta = \frac{u_r}{r}, \gamma_{r\theta} = 0 \quad (15)$$

one can obtain the differential equation of radial displacement

$$\frac{d^2 u_r}{dr^2} + \frac{1}{r} \frac{du_r}{dr} - \frac{u_r}{r^2} = \frac{1+\mu}{1-\mu} \frac{dT}{dr} \quad (16)$$

Let the distributed interfacial normal pressures be p_k ($k = 2, 3, \dots, N$), the radial displacement in k th layer of the GPLRC cylindrical shell can be obtained by solving Eq. (16)

$$u_{rk}(r) = \frac{1+\mu_k}{1-\mu_k} \frac{\alpha_k}{r} \int_{R_k}^r T(r) r dr + C_{1k} r + \frac{C_{2k}}{r} \quad (R_k \leq r \leq R_{k+1}, k = 1, 2, \dots, N) \quad (17)$$

where C_{1k} and C_{2k} can be determined by the pressure condition at the inner and outer surfaces of each layer, i.e.,

$$\begin{cases} C_{1k} = \frac{1}{\gamma_k} \frac{\varpi_k^2 p_k - p_{k+1} + p_{Tbk}}{1 - \varpi_k^2} \\ C_{2k} = \frac{1}{\eta_k} \frac{R_k^2 (p_k - p_{k+1} + p_{Tbk})}{1 - \varpi_k^2} \end{cases} \quad (k = 1, 2, \dots, N) \quad (18)$$

with

$$\begin{aligned} \varpi_k &= \frac{R_k}{R_{k+1}}, \quad \eta_k = \frac{E_k}{(1+\mu_k)}, \quad \gamma_k = \frac{E_k}{(1+\mu_k)(1-2\mu_k)}, \quad p_{Tbk} = \frac{\theta_k}{R_{k+1}^2} \int_{R_k}^{R_{k+1}} T_k(r) r dr, \\ \theta_k &= \frac{E_k \alpha_k}{1-\mu_k} \quad (k = 1, 2, \dots, N). \end{aligned} \quad (19)$$

Substituting Eq. (18) into Eq. (17) yields

$$u_{rk}(r) = \frac{1+\mu_k}{1-\mu_k} \frac{\alpha_k}{r} \int_{R_1}^r T_k(r) r dr + \frac{R_k^2/r + (1-2\mu_k)\varpi_k^2 r}{\eta_k(1-\varpi_k^2)} p_k - \frac{R_k^2/r + (1-2\mu_k)r}{\eta_k(1-\varpi_k^2)} p_{k+1} + \frac{R_k^2/r + (1-2\mu_k)r}{\eta_k(1-\varpi_k^2)} p_{Trk} \quad (20)$$

with

$$p_{Trk} = \frac{\theta_k}{r^2} \int_{R_k}^r T_k(r) r dr \quad (k = 1, 2, \dots, N). \quad (21)$$

The radial displacement continuity condition at the interface between the k th and $(k+1)$ -th layers requires

$$u_{bk} = u_{ak+1} \quad (k = 1, 2, \dots, N-1) \quad (22)$$

where u_{bk} and u_{ak+1} are the radial displacements of outer surface in the k th layer and inner surface in the $(k+1)$ th layer, respectively, and their expressions follow

$$u_{bk} = M_{Ak} p_k + M_{Bk} p_{k+1} + M_{Ck}, \quad u_{ak+1} = Q_{Ak} p_{k+1} + Q_{Bk} p_{k+2} + Q_{Ck} \quad (k = 1, 2, \dots, N-1) \quad (23)$$

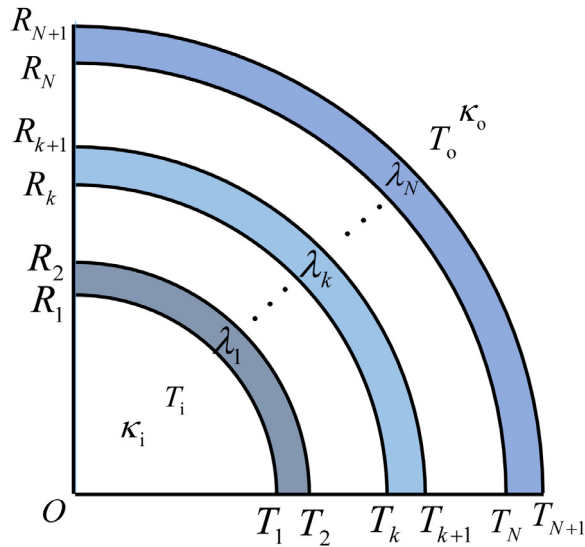


Fig. 4. Thermal transfer model of the hemispherical ends.

By substituting Eq. (23) into Eq. (22), one obtains

$$M_{Ak}p_k + (M_{Bk} - Q_{Ak})p_{k+1} - Q_{Bk}p_{k+2} = Q_{Ck} - M_{Ck} \quad (24)$$

with

$$M_{Ak} = \frac{2(1 - \mu_k)\varpi_k^2 R_{k+1}}{\eta_k(1 - \varpi_k^2)}, \quad M_{Bk} = \frac{-(\varpi_k^2 - 2\mu_k + 1)R_{k+1}}{\eta_k(1 - \varpi_k^2)}, \quad M_{Ck} = \frac{2(1 - \mu_k)R_{k+1}}{\eta_k(1 - \varpi_k^2)} p_{Tbk},$$

$$Q_{Ak} = \frac{[\varpi_{k+1}^2(1 - 2\mu_{k+1}) + 1]R_{k+1}}{\eta_{k+1}(1 - \varpi_{k+1}^2)}, \quad Q_{Bk} = \frac{-2(1 - \mu_{k+1})R_{k+1}}{\eta_{k+1}(1 - \varpi_{k+1}^2)}, \quad Q_{Ck} = \frac{2(1 - \mu_{k+1})R_{k+1}}{\eta_{k+1}(1 - \varpi_{k+1}^2)} p_{Tbk+1}. \quad (25)$$

For convenience, Eq. (24) can be expressed in the following matrix form as

$$\begin{bmatrix} M_{B1} - Q_{A1} & -Q_{B1} & 0 & \cdots & 0 \\ M_{A2} & M_{B2} - Q_{A2} & -Q_{B2} & \ddots & 0 \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -Q_{BN-1} \\ 0 & 0 & 0 & M_{AN-1} & M_{BN-1} - Q_{AN-1} \end{bmatrix} \begin{Bmatrix} p_2 \\ p_3 \\ \vdots \\ p_N \end{Bmatrix} = \begin{Bmatrix} Q_{C1} - M_{C1} - M_{A1}p_1 \\ Q_{C2} - M_{C2} \\ \vdots \\ Q_{CN} - M_{CN} \\ Q_{CN-1}M_{CN-1} + Q_{BN-1}p_{N+1} \end{Bmatrix} \quad (26)$$

Solving Eq. (26) to obtain $p_2, \dots, p_N$, and incorporating the prescribed pressures p_1 and p_{N+1} at the inner and outer surfaces, the coefficients C_{1k} and C_{2k} in each layer can be determined from Eq. (18), the radial stress $\sigma_{rk}(r)$, and hoop stress $\sigma_{\theta k}(r)$ in the k -th layer can be determined by

$$\begin{cases} \sigma_{rk}(r) = \gamma_k C_{1k} - \eta_k \frac{C_{2k}}{r^2} - p_{Trk} \\ \sigma_{\theta k}(r) = \gamma_k C_{1k} + \eta_k \frac{C_{2k}}{r^2} + p_{Trk} - \theta_k T_k(r) \end{cases} \quad (R_k \leq r \leq R_{k+1}, k = 1, 2, \dots, N) \quad (27)$$

3.2. Thermo-elastic analysis of hemispherical ends

The thermo-elastic analysis for the hemispherical ends in the closed FG-GPLRC cylindrical shell shown in Fig. 4 is carried out in this section. Under the same thermal boundary conditions discussed in Section 3.1, the steady-state thermal conductivity equation of the hemispherical ends is expressed as

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) T(r) = 0 \quad (28)$$

The general solution of Eq. (28) is

$$T(r) = D_k + \frac{F_k}{r} \quad (R_k \leq r \leq R_{k+1}, k = 1, 2, \dots, N) \quad (29)$$

where D_k and F_k are the constants determined by the thermal boundary conditions.

By introducing the continuous thermal flux condition along the radial direction below

$$F_k \lambda_k = F_{k+1} \lambda_{k+1} (k = 1, 2, \dots, N-1) \quad (30)$$

and considering Eqs. (29) and (30), the ratio between the interface temperature differences of adjacent layers can be written as

$$\frac{T_k - T_{k+1}}{T_{k+1} - T_{k+2}} = \xi_k (k = 1, 2, \dots, N-1) \quad (31)$$

where

$$\xi_k = \frac{\lambda_{k+1}}{\lambda_k} \frac{\frac{1}{R_k} - \frac{1}{R_{k+1}} + \omega_1 \frac{\lambda_k}{R_k^2 \kappa_i}}{\frac{1}{R_{k+1}} - \frac{1}{R_{k+2}} + \omega_2 \frac{\lambda_{k+1}}{R_{k+2}^2 \kappa_o}} \quad (32)$$

Therefore, the temperature at each interface can be obtained as

$$\begin{bmatrix} 1 + \xi_1 & -\xi_1 & 0 & \cdots & 0 \\ -1 & 1 + \xi_2 & -\xi_2 & \ddots & 0 \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\xi_{N-2} \\ 0 & 0 & 0 & -1 & 1 + \xi_{N-1} \end{bmatrix} \begin{Bmatrix} T_2 \\ T_3 \\ \vdots \\ T_N \end{Bmatrix} = \begin{Bmatrix} T_1 \\ 0 \\ \vdots \\ 0 \\ \xi_{N-1} T_{N+1} \end{Bmatrix} \quad (33)$$

and so D_k and F_k are determined as

$$\begin{cases} D_k = \frac{T_{k+1} R_{k+1} - T_k R_k + \omega_1 T_{k+1} R_{k+1} \frac{\lambda_k}{R_k \kappa_i} + \omega_2 T_k R_k \frac{\lambda_k}{R_{k+1} \kappa_o}}{(R_{k+1} - R_k) + \omega_1 R_{k+1} \frac{\lambda_k}{R_k \kappa_i} + \omega_2 R_k \frac{\lambda_k}{R_{k+1} \kappa_o}} \\ F_k = \frac{-R_k R_{k+1} (T_{k+1} - T_k)}{(R_{k+1} - R_k) + \omega_1 R_{k+1} \frac{\lambda_k}{R_k \kappa_i} + \omega_2 R_k \frac{\lambda_k}{R_{k+1} \kappa_o}} \end{cases} (k = 1, 2, \dots, N) \quad (34)$$

in which ω_1 and ω_2 are constants distinguishing the prescribed thermal boundary conditions.

It is assumed that w_{bk} and w_{ak+1} are the radial displacements of outer surface in the k -th layer and inner surface in the $(k+1)$ th layer of the hemispherical ends, respectively. The displacement continuity condition of the hemispherical ends can be written as

$$w_{bk} = w_{ak+1} (k = 1, 2, \dots, N-1) \quad (35)$$

with

$$w_{bk} = M_{Dk} p_k + M_{Ek} p_{k+1} + M_{Fk}, \quad w_{ak+1} = Q_{Dk} p_{k+1} + Q_{Ek} p_{k+2} + Q_{Fk} \quad (36)$$

Substituting Eq. (36) into Eq. (35) yields

$$M_{Dk} p_k + (M_{Ek} - Q_{Dk}) p_{k+1} - Q_{Ek} p_{k+2} = Q_{Fk} - M_{Fk} (k = 1, 2, \dots, N-1) \quad (37)$$

with

$$\begin{aligned} M_{Dk} &= \frac{3 \frac{1-\mu_k}{1+\mu_k} \varpi_k^3 R_{k+1}}{2\eta_k (1 - \varpi_k^3)}, \quad M_{Ek} = -\frac{(2 \frac{1-2\mu_k}{1+\mu_k} + \varpi_k^3) R_{k+1}}{2\eta_k (1 - \varpi_k^3)}, \quad M_{Fk} = \frac{3 \frac{1-\mu_k}{1+\mu_k} R_{k+1}}{\eta_k (1 - \varpi_k^3)} p_{Sbk}, \\ V_{Dk} &= \frac{(2 \frac{1-2\mu_{k+1}}{1+\mu_{k+1}} \varpi_{k+1}^3 + 1) R_{k+1}}{2\eta_{k+1} (1 - \varpi_{k+1}^3)}, \quad V_{Ek} = -\frac{3 \frac{1-\mu_{k+1}}{1+\mu_{k+1}} R_{k+1}}{2\eta_{k+1} (1 - \varpi_{k+1}^3)}, \quad V_{Fk} = \frac{3 \frac{1-\mu_{k+1}}{1+\mu_{k+1}} R_{k+1}}{\eta_{k+1} (1 - \varpi_{k+1}^3)} p_{Sbk+1}, \\ p_{Sbk} &= \frac{\theta_k}{R_{k+1}^3} \int_{R_k}^{R_{k+1}} T_k(r) r^2 dr (k = 1, 2, \dots, N). \end{aligned} \quad (38)$$

and the matrix form of Eq. (37) is

$$\begin{bmatrix} M_{E1} - Q_{D1} & -Q_{E1} & 0 & \cdots & 0 \\ M_{D2} & M_{E2} - Q_{D2} & -Q_{E2} & \ddots & 0 \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -Q_{EN-1} \\ 0 & 0 & 0 & M_{DN-1} & M_{EN-1} - Q_{EN-1} \end{bmatrix} \begin{Bmatrix} p_2 \\ p_3 \\ \vdots \\ p_N \end{Bmatrix} = \begin{Bmatrix} Q_{F1} - M_{F1} - M_{D1} p_1 \\ Q_{F2} - M_{F2} \\ \vdots \\ Q_{FN} - M_{FN} \\ V_{FN-1} - M_{FN-1} + V_{EN-1} p_{N+1} \end{Bmatrix} \quad (39)$$

Table 1

Material properties and geometrical parameters of each layer in the composite cylindrical shell.

	λ / W/(m.K)	E /GPa	α_G / 10^{-6} /K	ν	Thickness δ / mm
1st layer	18.4	190	18.7	0.28	10
2nd layer	25.4	194	17.4	0.28	10
3rd layer	32.4	198	16.1	0.28	10
4th layer	39.4	202	14.8	0.28	10
5th layer	46.4	206	13.5	0.28	10
6th layer	53.4	210	12.2	0.30	30

The coefficients C_{3k} and C_{4k} are determined as

$$\begin{cases} C_{3k} = \frac{\varpi_k^3 p_k - p_{k+1} + 2p_{Sbk}}{\chi_k(1 - \varpi_k^3)} \\ C_{4k} = \frac{R_k^3(p_k - p_{k+1} + 2p_{Sbk})}{2\eta_k(1 - \varpi_k^3)} \end{cases} \quad (k = 1, 2, \dots, N) \quad (40)$$

with

$$\chi_k = \frac{E_k}{1 - 2\mu_k} \quad (41)$$

Therefore, the radial displacement $u_{rk}(r)$, radial stress $\sigma_{rk}(r)$ and hoop stress $\sigma_{\theta k}(r)$ of the hemisphere ends can be obtained as

$$\begin{cases} u_{rk}(r) = \frac{1 + \mu_k}{1 - \mu_k} \frac{\alpha_k}{r^2} \int_{R_k}^{R_{k+1}} T_k(r) r^2 dr + C_{k3} r + \frac{C_{k4}}{r^2} \\ \sigma_{rk}(r) = \chi_k C_{k3} - 2\eta_k C_{k4}/r^3 - 2p_{Srk} \\ \sigma_{\theta k}(r) = \sigma_{\varphi k} = \chi_k C_{k3} + \eta_k C_{k4}/r^3 + p_{Srk} - \theta_k T_k \end{cases} \quad (R_k \leq r \leq R_{k+1}, k = 1, 2, \dots, N) \quad (42)$$

where

$$p_{Srk} = \frac{\theta_k}{r^3} \int_{R_k}^r T_k(r) r^2 dr \quad (43)$$

The temperatures and radial pressures at the intermediate interfaces of the FG-GPLRC cylindrical shell satisfy Eqs. (12) and (26) and the temperatures and radial pressures of the intermediate interfaces of the hemispherical ends satisfy Eqs. (33) and (39). All of them are $(N-1)$ order linear equations and can be calculated numerically through the following process:

- (1) Determine parameters $T_1, T_{N+1}, T_1, T_0, \kappa_0, T_i, \kappa_i, T_{N+1}, T_i, T_0, \kappa_i, \kappa_0$ for given thermal boundary conditions;
- (2) Calculate β_k and ξ_k ;
- (3) Calculate T_k in the FG-GPLRC cylindrical section and hemispherical ends;
- (4) Calculate A_k, B_k, C_k, D_k and determine the temperature distribution;
- (5) Calculate the interlayer normal pressure p_k in the FG-GPLRC cylindrical section and hemispherical ends;
- (6) Calculate the stress coefficients C_{1k}, C_{2k}, C_{3k} and C_{4k} ;
- (7) Determine the radial displacement, radial and hoop stresses of the FG-GPLRC cylindrical and hemispherical shells.

4. Results and discussions

Thermo-elastic responses of the closed FG-GPLRC composite cylindrical shell are discussed through numerical results in this section. The inner radius and thickness of the shell are 200 and 7.2 mm, and the length of the cylindrical part is 2 m. In what follows, the material properties of the polymer matrix and GPLs are [37]: $E_M = 2.85$ GPa, $\nu_M = 0.38$, $\alpha_M = 60 \times 10^{-6}$ /K, $\lambda_M = 5.8$ W/(m.K), $c_M = 1110$ J/(kg.K), $\rho_M = 1.2$ g/cm³, $E_G = 1.01$ TPa, $\nu_G = 0.2$, $\alpha_G = 5.0 \times 10^{-6}$ /K, $\lambda_G = 3000$ W/(m.K), $c_G = 644$ J/(kg.K), $\rho_G = 1.06$ g/cm³, $l_G = 2.5$ μ m, $w_G = 1.5$ μ m, $h_G = 1.5$ nm, $f_G = 0.5\%$, and $\kappa_i = \kappa_0 = 15$ W/(m².K), unless otherwise specified.

4.1. Convergence and validation

To validate the present analysis, the radial and hoop stresses a 6-layer closed composite cylindrical shell with parameters listed in Table 1 are compared with those in Ref. [44] in Fig. 5. The shell is under the internal pressure 22 MPa and external pressure 1.5 MPa. The temperature at the inner and outer surfaces of the composite cylinder are fixed as 200 °C and 150 °C, respectively. Our results are in good agreement with those reported in Ref. [44]. Note that the curves marked with “ $\times$ ” and

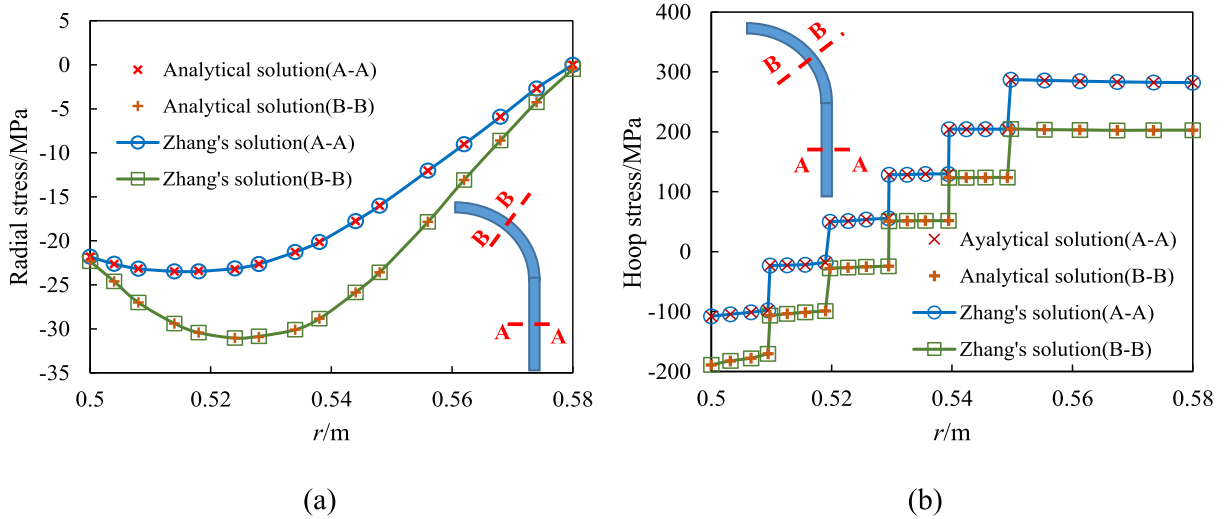


Fig. 5. Radial and hoop stresses of the 6-layer closed composite cylindrical shell.

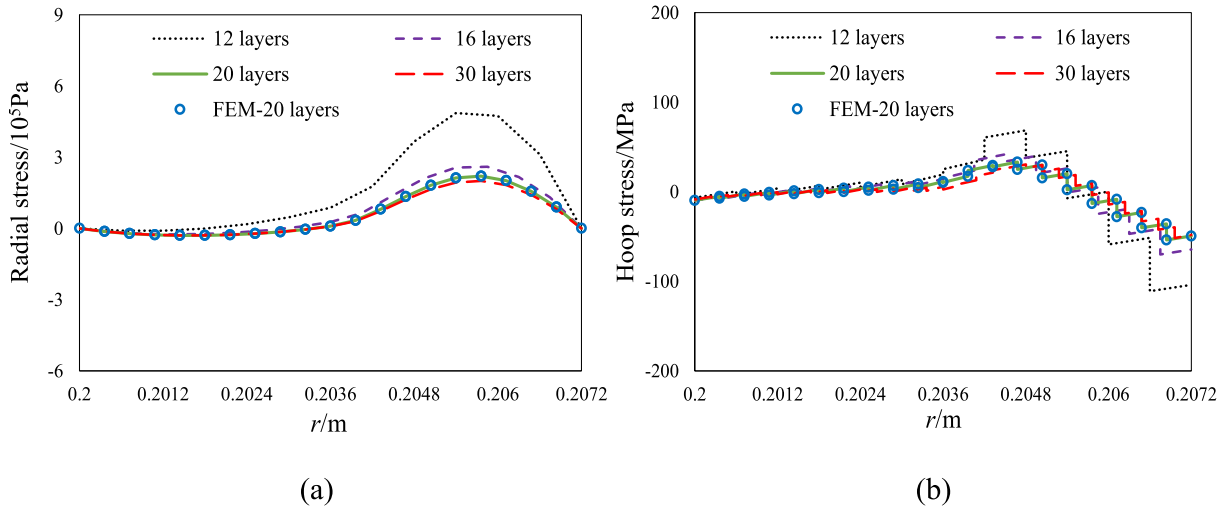


Fig. 6. Thermal stresses of the FG-GPLRC cylindrical shell: (a) radial stress, (b) hoop stress.

“+” denote the stresses along the cross-section A-A in the cylindrical body and the cross-section B-B in the hemispherical end.

Figs. 6 and 7 plot the radial stress and hoop stress of the FG-X-GPLRC cylindrical shell and hemispherical ends with different number of GPLRC layers under TBC 1, where the external pressure p_1 and p_{N+1} are set to zero, and the temperatures at the internal and external surfaces are fixed at 200 °C and 20 °C, respectively. It is observed that the relative difference between the results with $N = 20$ and $N = 30$ is quite small. Hence, $N = 20$ is used in all of the following numerical results.

Moreover, finite element results using ABAQUS 6.14 are also plotted in Figs. 6 and 7 for further comparison. As shown in Fig. 8, the closed FG-GPLRC cylindrical shell is meshed into 119,860 elements including 8-node linear heat transfer brick DC3D8 elements and 8-node coupled temperature-displacement C3D8T elements. Again, good agreement is observed in both Figs. 6 and 7.

4.2. Thermo-elastic responses

Fig. 9 shows the temperature variation along the thickness of the closed FG-GPLRC cylindrical shell under TBC 1. It is found that the temperature variation of the closed FG-U-GPLRC cylindrical shell is linear. For the closed FG-X-GPLRC and FG-O-GPLRC cylindrical shells, their temperature variation curves are S-shaped while those of the closed FG-V-GPLRC and FG-A-GPLRC cylindrical shell are convex and concave, respectively. This is due to the fact that the GPLRC layer with higher GPL concentration has a higher thermal conductivity consequently leading to more heat transfer.

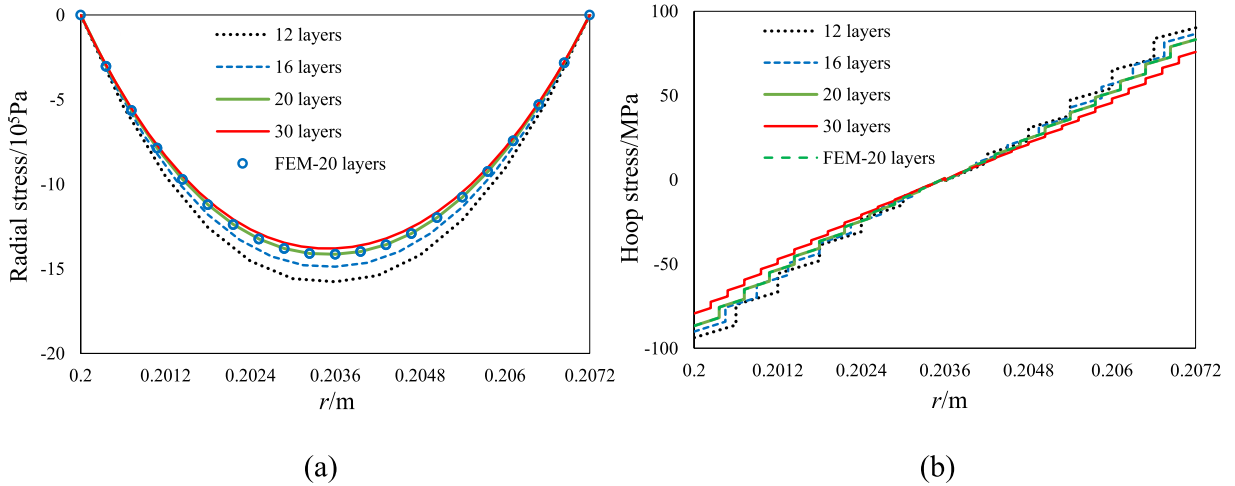


Fig. 7. Thermal stresses at the FG-GPLRC hemispherical ends: (a) radial stress, (b) hoop stress.

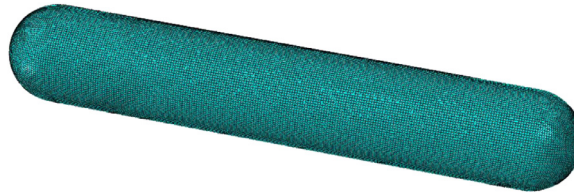


Fig. 8. FE model of the closed FG-GPLRC cylindrical shell.

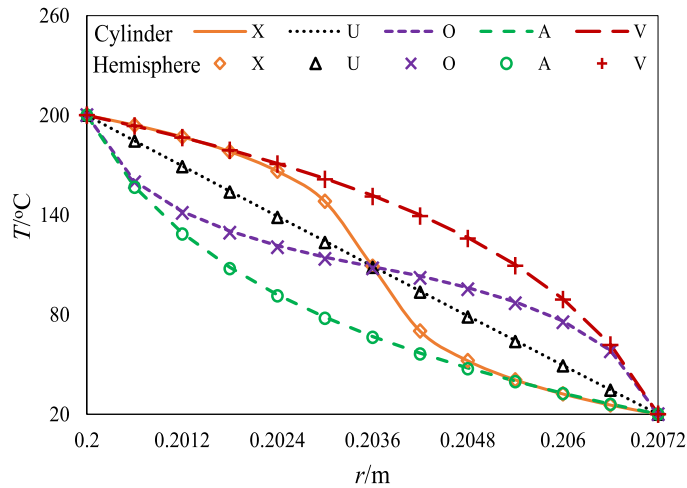


Fig. 9. Temperature variation of a closed FG-GPLRC cylindrical shell.

Figs. 10 and 11 plot the radial stress and hoop stress along the thickness direction of the FG-GPLRC cylindrical shell and hemispherical end under TBC 1, respectively. It should be noted from Fig. 10a that the radial stresses in both FG-U-GPLRC and FG-V-GPLRC cylindrical shells are compressive while those in FG-X-GPLRC, FG-O-GPLRC and FG-A-GPLRC cylindrical shells are tensile. The hoop stress shown in Fig. 10b is quite complicated for FG-GPLRC cylindrical shells under TBC 1. In particular, the radial stress and hoop stress of the FG-V-GPLRC cylindrical shell are the smallest among the five GPL distribution patterns considered.

The radial and hoop stresses at the hemispherical end are displayed in Fig. 11a. It can be seen that the entire hemispherical ends are subjected to compressive radial stress while the hoop stress varies from compressive to tensile along the thickness direction. In addition, the hemispherical end with GPL distribution pattern O has the smallest radial and hoop

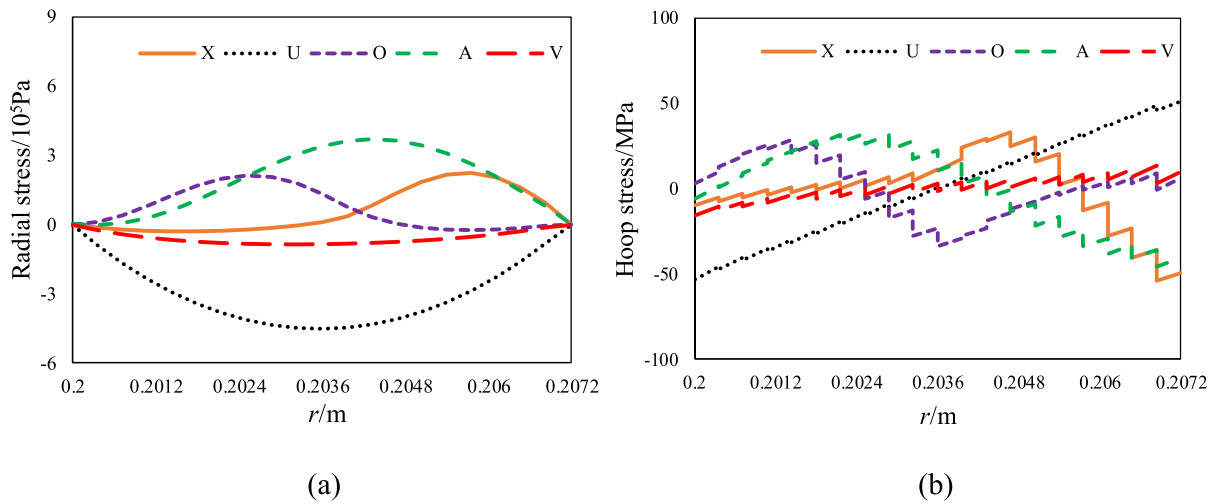


Fig. 10. (a) Radial stress and (b) hoop stress of the FG-GPLRC cylindrical shell.

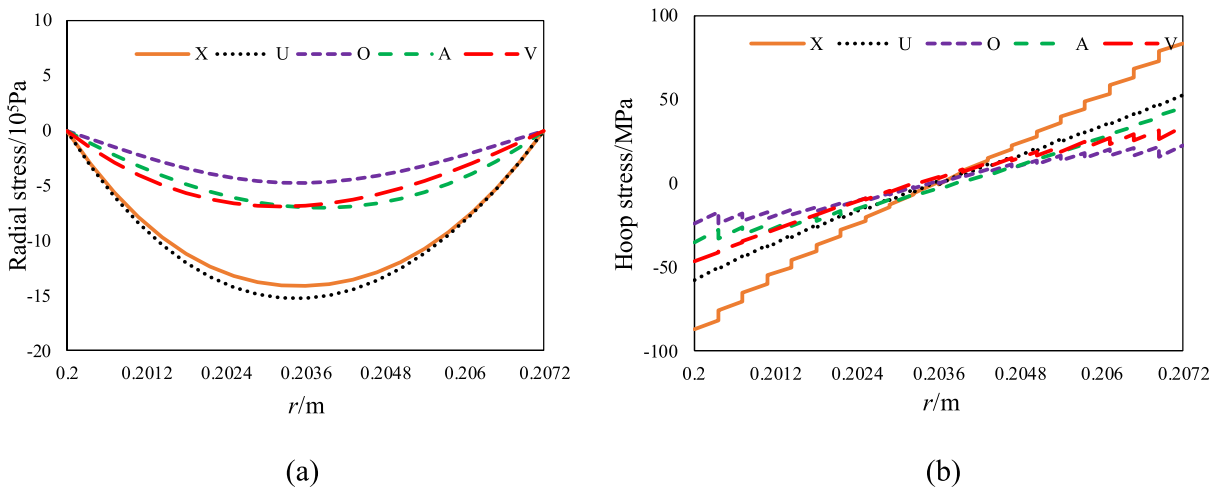


Fig. 11. (a) Radial stress and (b) hoop stress at the hemispherical ends.

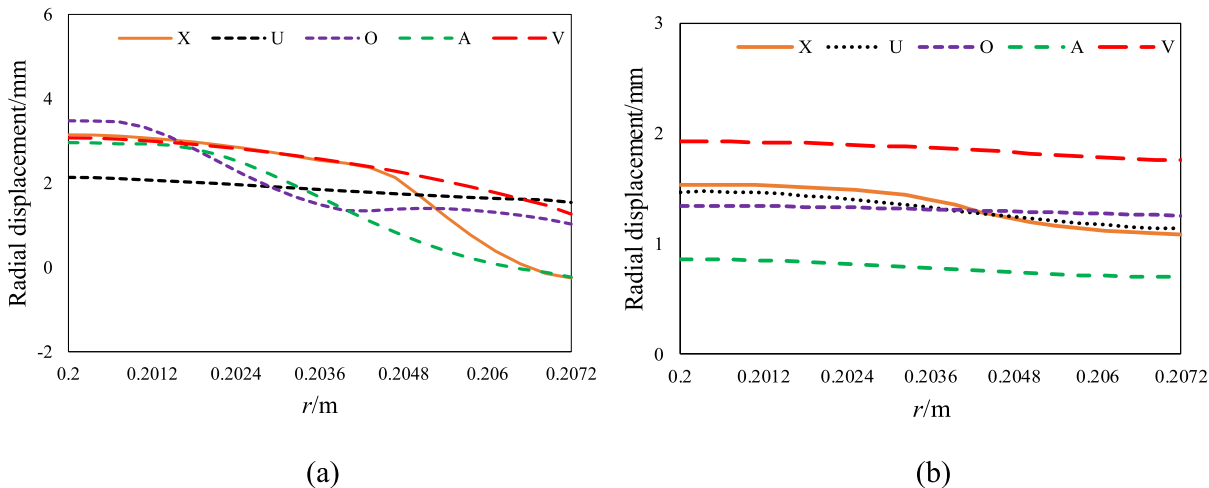


Fig. 12. Radial displacement of (a) the FG-GPLRC cylindrical shell and (b) hemispherical end.

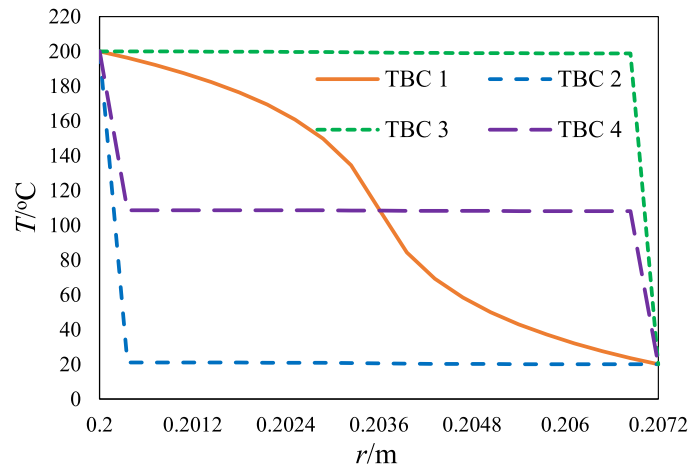


Fig. 13. Temperature distribution of the closed FG-X-GPLRC cylindrical shell under different TBCs.

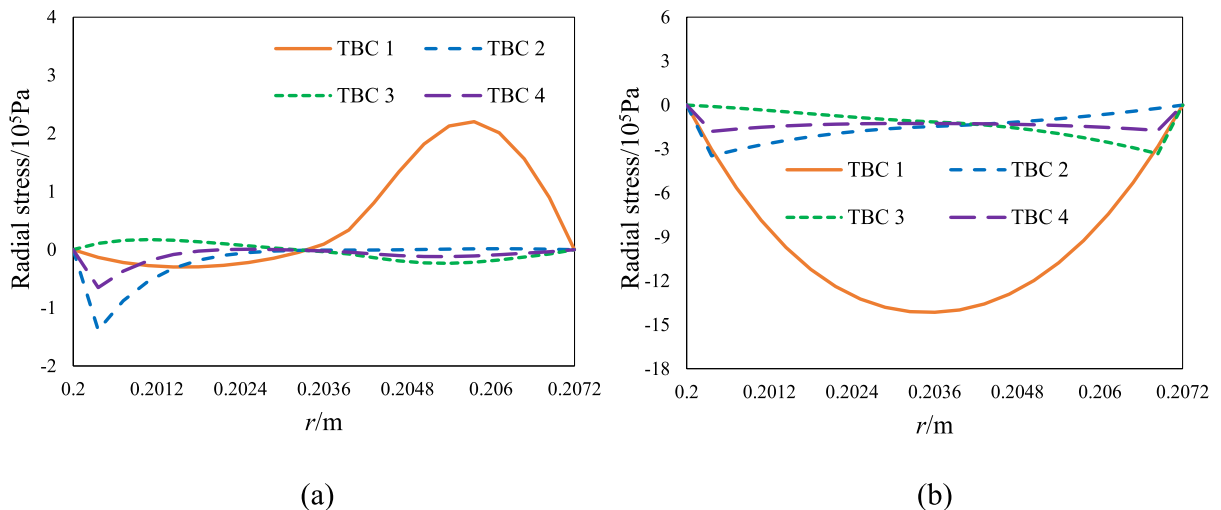


Fig. 14. Radial stress of the closed FG-X-GPLRC shell under different TBCs: (a) the cylindrical section, and (b) hemispherical ends.

stresses. Fig. 12 gives the radial displacement of the FG-GPLRC cylindrical section and hemispherical ends under TBC 1, showing the significant influence of the thermal effect on the closed FG-GPLRC cylindrical shell.

Fig. 13 shows the temperature distribution along the thickness direction of the closed FG-X-GPLRC cylindrical shell under four TBCs where the temperatures at the inner and outer surfaces are fixed at 200 °C and 20 °C, respectively. In all cases, the shell undergoes a significant thermal gradient along its thickness direction although the temperature distribution is quite different. It should be noted that TBC 1 leads to a temperature field completely different from those under other TBCs that exhibit a huge temperature gradient near the inner and outer surfaces only but remain constant in other regions. As a result, the radial stress distribution in the closed FG-X-GPLRC cylindrical shell under TBC 1 (as shown in Fig. 14) is remarkably different from and has a higher peak value than the results with other TBCs. The hoop stress of the same closed shell is depicted in Fig. 15. The maximum hoop stress is found at the convection interface when the closed shell is under TBC 2, TBC 3 and TBC 4.

Fig. 16 demonstrates the effect of GPL weight fraction on the temperature distribution, radial stress, and hoop stress of the closed FG-X-GPLRC cylindrical shell. It can be observed from Fig. 16a that an increase in GPL weight fraction leads to higher thermal conductivity of the GPLRC layer and consequently dramatic heat transfer among GPLRDC layers, which in turn results in a significant change in the thermally induced radial stress and hoop stress in the closed FG-X-GPLRC cylindrical shell, as shown in Fig. 16b and c.

Figs. 17 and 18 discuss the influences of the GPL geometry and dimension on the radial stress and hoop stress in the FG-X-GPLRC cylindrical section and hemispherical ends under TBC 1, respectively. Numerical results show that the effects of GPL geometry and dimension on the radial and hoop stresses of the closed FG-X-GPLRC cylindrical shell are relatively limited.

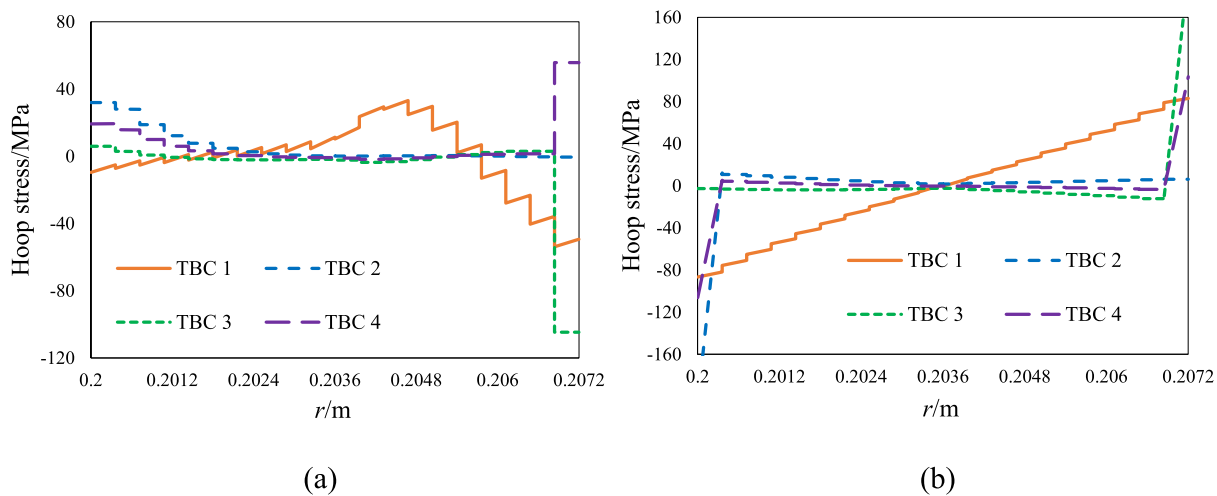


Fig. 15. Hoop stress of (a) the FG-X-GPLRC cylindrical shell and (b) hemispherical end under different TBCs.

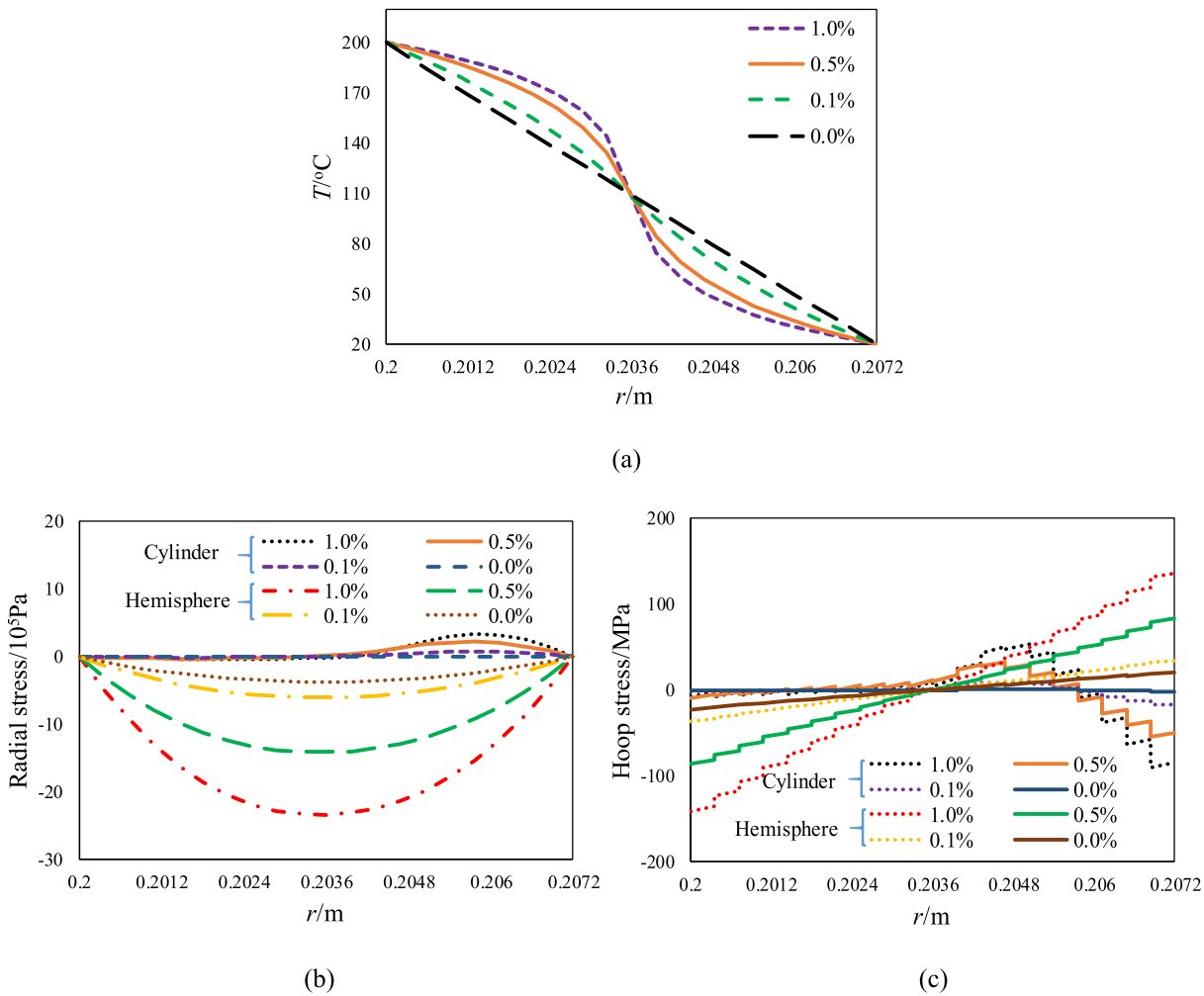


Fig. 16. Effect of GPL weight fraction on the (a) temperature distribution, (b) radial stress, and (c) hoop stress of the closed FG-X-GPLRC cylindrical shell.

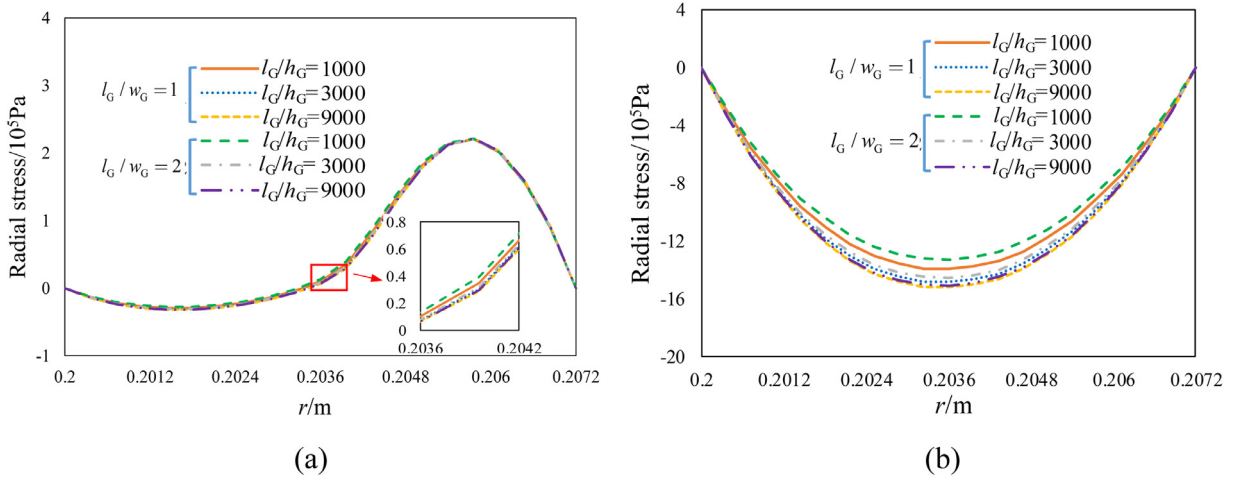


Fig. 17. Effects of GPL geometry and dimension on radial stress of the closed FG-X-GPLRC cylindrical shell: (a) cylindrical section, and (b) hemispherical ends.

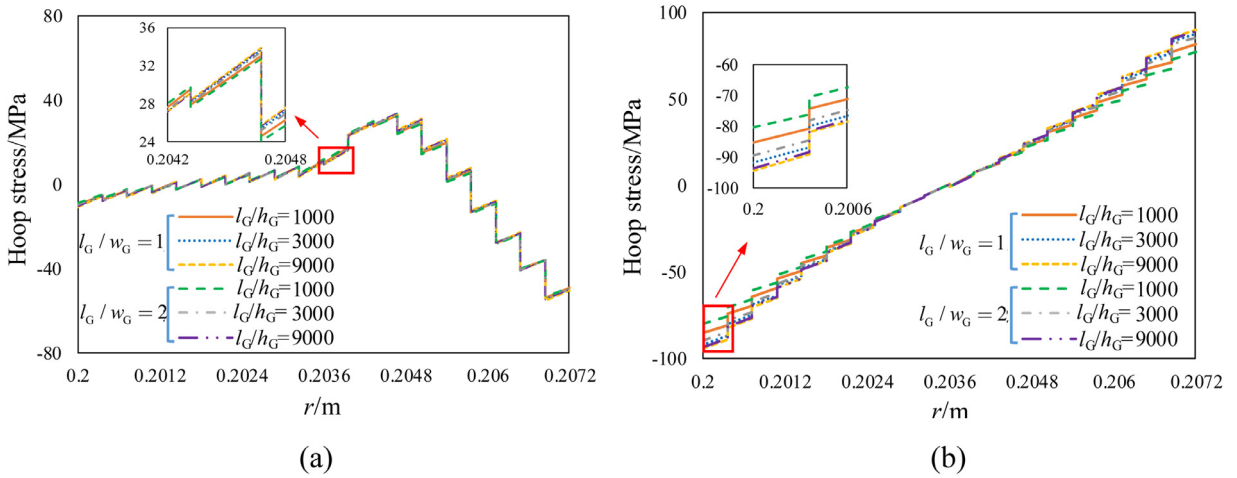


Fig. 18. Effects of GPL geometry and dimension on hoop stress of the closed FG-X-GPLRC cylindrical shell: (a) cylindrical section, and (b) hemispherical ends.

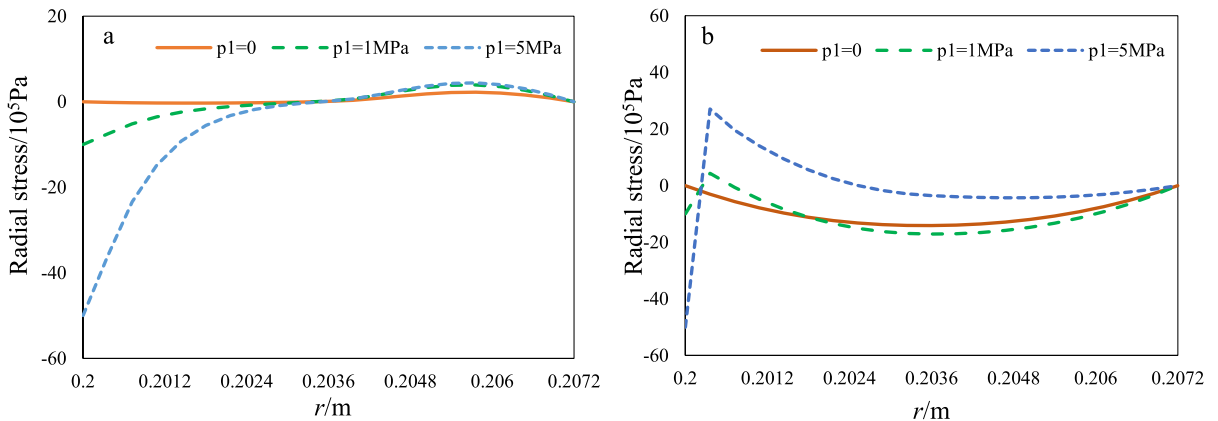


Fig. 19. Effect of internal pressure on radial stress of closed FG-X-GPLRC cylindrical shell under TBC 1: (a) cylindrical section; and (b) hemispherical ends.

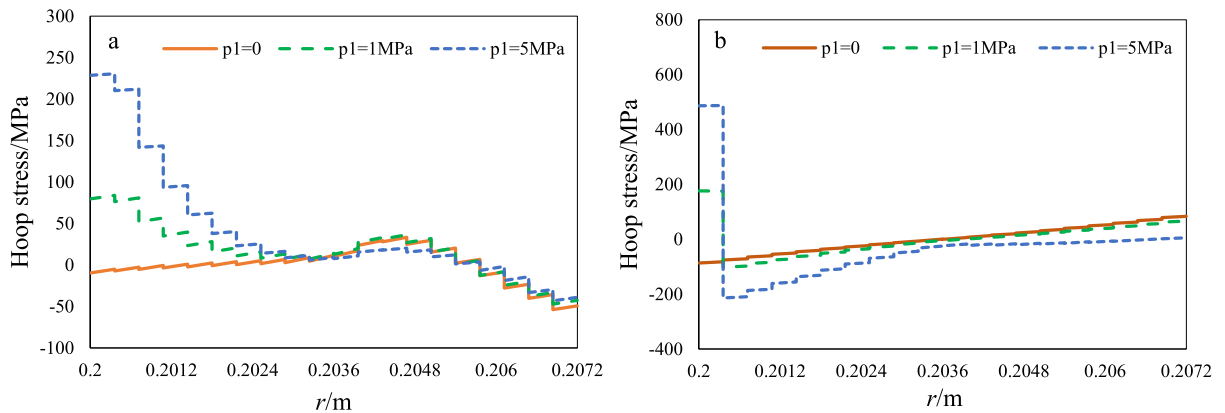


Fig. 20. Effect of internal pressure on hoop stress of closed FG-X-GPLRC cylindrical shell under TBC 1: (a) cylindrical section; and (b) hemispherical ends.

Figs. 19 and 20 show the influences of the internal pressure on the radial stress and hoop stress in the FG-X-GPLRC cylindrical section and hemispherical ends under TBC 1, respectively. The total radial and hoop stresses in the cylindrical section and hemispherical ends are the superposition results of stresses caused by thermal load and internal pressure alone. As shown from Fig. 19(a), internal pressure increases interlayer radial pressure in inner half of the cylindrical section, which significantly improve the hoop stresses at these regions. Under fixed temperature condition, each layer in the hemispherical end is under radial pressure, however, the existence internal pressure causes a significant inflection point of radial stress at the outer surface of 1-st layer, results in the large jump of hoop stress at the interface of 1-st and 2-nd layers in the hemispherical end.

5. Conclusions

In this paper, a comprehensive thermo-elastic model is established for closed FG-GPLRC cylindrical shells under a thermal loading and its thermally induced displacement and stress behaviors due to through-thickness steady-state thermal conduction with different thermal boundary conditions are investigated for the first time. The influences of GPL distribution pattern, weight fraction, dimension as well as thermal boundary conditions (TBCs) on the thermal radial stress, hoop stress and displacement have been discussed in detail. It is concluded that

- (1) GPL distribution patterns V and O are the preferred pattern for FG-GPLRC cylindrical section and hemispherical ends, respectively.
- (2) The closed FG-X-GPLRC cylindrical shell with a higher GPL concentration has a higher thermal conductivity consequently has higher thermal stresses.
- (3) GPL geometry and dimension have relatively limited influences on the radial stress and hoop stress of the closed FG-X-GPLRC cylindrical shell.
- (4) The results and research findings obtained in the present study are of practical importance for the applications and design of closed FG-GPLRC shells in a variety of engineering sectors including aerospace propeller, nuclear reactors and chemical plants, etc. The proposed model can also be readily generalized to deal with the thermo-electro-elastic and thermal radiation responses of multi-layered cylindrical shell made of FG-GPLRC as well as other composite materials.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Xiaoguang Huang: Conceptualization, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Funding acquisition. **Jie Yang:** Methodology, Resources, Supervision, Writing – review & editing. **Zhicheng Yang:** Investigation, Data curation, Validation.

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